

VANDENBERG AFB, CA, USA

WMO: 723930

Lat: 34.717N

Lon: 120.567W

Elev: 369

StdP: 14.50

Time Zone: -8.00 (NAP)

Period: 95-19

WBAN: 93214

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	35.7	37.2	20.7	15.7	51.8	24.7	19.0	51.0	27.7	56.0	25.7	55.8	6.9	110	0.550

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	16.1	80.6	60.3	75.7	59.2	72.5	59.0	65.0	72.9	63.4	70.2	62.2	68.5	8.3	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.5	85.7	66.5	60.9	81.0	65.2	59.5	76.9	64.1	30.1	72.4	28.9	70.4	28.0	68.5	77.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.3	23.5	20.1	DB	30.2	91.6	3.6	5.7	27.6	95.7	25.5	99.1	23.5	102.3	20.9	106.5	(4)
(5)			WB	27.5	68.5	3.3	3.3	25.1	70.8	23.2	72.8	21.3	74.6	18.9	77.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	56.2	52.2	52.5	53.3	54.1	55.6	57.1	59.8	60.5	61.1	59.8	56.2	52.0		
	DBStd	5.36	5.01	5.19	4.13	4.27	3.50	3.10	2.78	2.82	4.08	5.04	5.13	4.72		
	HDD50	117	33	27	12	8	0	0	0	0	0	0	5	32		
	HDD65	3265	397	351	362	331	295	240	165	142	132	178	269	403		
	CDD50	2385	102	98	116	130	173	213	303	327	333	304	192	94		
	CDD65	56	0	1	1	2	2	3	3	4	16	18	6	0		
	CDH74	607	10	25	19	37	43	38	10	26	123	200	71	5		
(13)	CDH80	172	0	4	3	10	16	19	2	5	36	62	15	0		
(14)	Wind	WSAvg	8.8	8.9	9.5	9.6	10.8	10.5	9.2	7.5	7.2	7.2	8.2	9.0		
(15) Precipitation	PrecAvg	16.6	3.6	3.7	2.7	1.1	0.4	0.1	0.0	0.0	0.1	0.8	1.3	2.8		
	PrecMax	32.6	16.3	15.7	12.6	4.3	2.4	0.7	0.3	0.1	0.4	4.0	4.5	9.3		
	PrecMin	6.2	0.2	0.1	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.2		
	PrecStd	6.6	3.5	3.3	2.8	1.1	0.6	0.2	0.1	0.0	0.1	0.9	1.1	2.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.3	78.0	76.5	79.0	79.3	77.3	74.3	77.0	85.1	87.3	81.4	73.2		
		MCWB	54.4	57.8	57.9	58.9	58.5	61.6	63.6	64.6	63.8	61.0	58.6	53.4		
	2%	DB	69.9	71.1	69.6	70.3	69.8	69.3	70.8	71.8	77.2	79.4	74.6	67.9		
		MCWB	53.2	54.7	54.9	55.7	58.9	59.8	62.4	62.5	63.0	59.5	56.6	53.3		
	5%	DB	64.9	65.2	64.7	64.8	65.0	65.8	68.1	69.2	72.5	73.5	69.4	63.7		
		MCWB	53.1	54.2	54.5	55.7	57.1	58.7	61.1	61.6	62.1	58.5	55.8	53.1		
	10%	DB	61.0	61.3	61.5	61.9	62.6	63.5	65.9	66.7	69.4	69.4	65.4	60.9		
		MCWB	53.1	53.2	54.1	54.5	55.9	57.3	59.9	60.5	61.3	58.8	55.5	53.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.1	60.4	60.2	60.9	62.3	63.2	65.5	65.8	68.2	66.8	62.1	60.7		
		MCDB	63.6	67.2	68.6	70.8	71.2	75.3	71.7	74.1	77.9	75.5	71.0	64.2		
	2%	WB	57.6	58.3	58.3	58.4	60.0	60.6	63.1	63.4	65.3	63.8	60.1	58.8		
		MCDB	61.2	64.4	64.9	66.2	67.0	67.3	69.4	69.6	72.9	70.2	67.4	61.8		
	5%	WB	56.3	56.5	56.6	56.7	58.0	59.2	61.5	62.1	63.4	62.0	58.6	56.8		
		MCDB	60.2	62.4	62.3	63.4	64.2	64.7	66.7	67.7	70.0	68.5	65.1	59.8		
	10%	WB	54.7	54.9	55.0	55.2	56.3	57.9	60.2	60.9	61.9	60.2	57.2	54.8		
		MCDB	58.9	60.4	60.0	60.8	61.7	62.6	64.9	65.9	67.9	66.7	63.2	58.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.7	18.0	17.3	16.6	14.6	13.4	12.6	13.4	16.1	19.2	19.6	18.5		
		MCDBR	25.0	25.5	24.2	23.2	20.7	18.6	16.2	16.7	22.5	27.3	26.3	23.2		
		MCWBR	13.7	13.7	13.1	12.7	11.8	10.7	8.6	8.8	11.6	14.0	14.0	14.0		
	5% WB	MCDBR	17.1	19.1	18.9	19.7	18.9	16.3	15.2	15.8	19.2	20.9	20.3	15.0		
		MCWBR	11.4	11.7	11.7	11.7	11.1	9.6	8.3	8.5	10.6	11.9	12.4	11.0		
(40) Clear-Sky Solar Irradiance	taub		0.302	0.316	0.323	0.339	0.349	0.349	0.357	0.342	0.343	0.335	0.304	0.306		
	taud		2.603	2.571	2.552	2.493	2.486	2.479	2.513	2.528	2.528	2.553	2.611	2.603		
	Ebn at Noon		290	297	302	300	296	295	292	296	292	286	287	282		
	Edh at Noon		24	28	30	34	35	35	34	32	31	28	23	23		
(44) All-Sky Solar Radiation	RadAvg		955	1227	1668	2076	2294	2354	2215	2051	1789	1389	1066	851		
	RadStd		73	123	107	128	152	111	84	97	98	105	60	75		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (26 neighbors)	+1.17	N/A	-2.45	+1.44	N/A	N/A	N/A	-301	+432	+137

Nomenclature: See separate page